

Table S1

Mapping brain regions to cortical segmentation in the Desikan atlas and subcortical segmentation in FreeSurfer

Model	Brain Region	Desikan Atlas / FreeSurfer Region
Cognitive control system	Dorsolateral Prefrontal Cortex (dlPFC)	Middle frontal gyrus
	Anterior Cingulate Cortex (ACC)	Caudal anterior cingulate cortex Rostral anterior cingulate cortex
Socio-emotional system	Orbitofrontal Cortex (OFC)	Medial orbitofrontal cortex Lateral orbitofrontal cortex
	Medial Prefrontal Cortex (mPFC)	Medial orbitofrontal cortex Superior frontal gyrus
	Amygdala	Amygdala
	Ventral Striatum (VS)	Nucleus accumbens
Default Mode Network	Dorsal Medial Prefrontal Cortex (dmPFC)	Superior frontal gyrus (medial)
	Anterior Medial Prefrontal Cortex (amPFC)	Medial orbitofrontal cortex
	Ventral Medial Prefrontal Cortex (vmPFC)	Medial orbitofrontal cortex
	Posterior Cingulate Cortex (PCC)	Posterior cingulate cortex
	Precuneus	Precuneus
	Lateral Temporal Cortex (LTC)	Middle temporal gyrus Superior temporal gyrus
	Inferior Parietal Lobule (IPL)	Inferior parietal cortex Supramarginal gyrus
	Retrosplenial Cortex (Rsp)	Isthmus cingulate cortex
	Hippocampal Formation (HF)	Hippocampus
	Parahippocampal Cortex (PHC)	Parahippocampal gyrus
	Angular Gyrus	Angular gyrus
	Temporal Pole	Temporal pole
	Superior Frontal Gyrus	Superior frontal gyrus
	Inferior Frontal Gyrus	Inferior frontal gyrus (pars triangularis, pars opercularis, pars orbitalis)
	Striatum (Caudate & Posterior Putamen)	Caudate nucleus Putamen)
	Cerebellum	Cerebellum
Salience Network	Anterior Insula (AI)	Insula
	Dorsal Anterior Cingulate Cortex (dACC)	Caudal anterior cingulate cortex
	Right Fronto-Insular Cortex (rFIC)	Insula Inferior frontal gyrus (pars opercularis)
	Ventral Striatum (VS)	Nucleus accumbens
	Thalamus	Thalamus

Table S2*Neural domains: ROIs and number of predictors*

System	Brain Regions	Domain	Predictors	Contrasts
Cognitive Control System	Dorsolateral Prefrontal Cortex (dlPFC) *	GMV	8	-
	Anterior Cingulate Cortex (ACC) *	CT	8	-
		rsfMRI	8	-
		fMRI SST	56	7
		fMRI MID	80	10
		fMRI nback	72	9
Socio-emotional System	Orbitofrontal Cortex (OFC) *	GMV	10	-
	Medial Prefrontal Cortex (mPFC) *	CT	6	-
	Amygdala	rsfMRI	10	-
	Ventral Striatum (VS) *	fMRI SST	70	7
		fMRI MID	100	10
		fMRI nback	90	9
Default Mode Network	Dorsal Medial Prefrontal Cortex (dmPFC) *	GMV	38	-
	Anterior Medial Prefrontal Cortex (amPFC)	CT	28	-
	Ventral Medial Prefrontal Cortex (vmPFC) *	rsfMRI	38	-
	Posterior Cingulate Cortex (PCC) *	fMRI SST	266	7
	Precuneus *	fMRI MID	380	10
	Lateral Temporal Cortex (LTC)	fMRI nback	342	9
	Inferior Parietal Lobule (IPL) *			
	Retrosplenial Cortex (Rsp)			
	Hippocampal Formation (HF) *			
	Parahippocampal Cortex (PHC) *			
	Angular Gyrus *			
	Temporal Pole *			
	Superior Frontal Gyrus *			
	Inferior Frontal Gyrus *			
	Striatum (Caudate & Posterior Putamen) *			
	Cerebellum			
Salience Network	Anterior Insula (AI) *	GMV	14	-
	Dorsal Anterior Cingulate Cortex (dACC) *	CT	10	-
	Right fronto-insular cortex (rFIC) *	rsfMRI	14	-
	Ventral Striatum (VS) *	fMRI SST	98	7
	Thalamus *	fMRI MID	140	10
		fMRI nback	126	9

Note. More information on contrasts in Table S4. *Dual system model:* list of brain regions based on (Steinberg et al., 2008), (Shulman et al., 2016), (Mennigen et al., 2014), & (Luna, Padmanabhan, & O’Hearn, 2010). *Default mode network:* list of brain regions based on (Buckner, Andrews-Hanna, & Schacter, 2008), (Andrews-Hanna, Reidler, Sepulcre, Poulin, & Buckner, 2010; Andrews-Hanna, Smallwood, & Spreng, 2014), & (Thomas Yeo et al., 2011). *Salience network:* list of brain regions based on (Seeley et al., 2007), (Uddin, 2015; Uddin, Supekar, Ryali, & Menon, 2011). GMV = Gray matter volume, CT = Cortical thickness, rsfMRI = Resting state fMRI, SST = Stop-signal task, MID = Monetary incentive delay, nback = emotional nBack task. The asterisk * marks brain regions and other predictors that have been found associated with delayed discounting in previous research.

Table S3*Predictors in cognitive and sociodemographic domains*

	Instrument	Cognitive skill	Measure
Cognitive skills	NIH Toolbox:		
	Pattern Comparison Processing Speed Test	Processing speed	Uncorrected stand. score
	Oral Reading Recognition Task	Reading decoding *	Uncorrected stand. score
	Picture Sequence Memory Test	Episodic memory	Uncorrected stand. score
	Flanker Inhibitory Control & Attention Task	Inhibition & Attention	Uncorrected stand. score
	Picture Vocabulary Test	Language comprehension*	Uncorrected stand. score
	Little Man Task	Visuospatial processing *	Efficiency score
	Rey Auditory Verbal Learning Test	Auditory learning, memory, & recognition	Average number of correctly memorized words across trials
		Characteristic	Source
Socio-demographics		Gender	Guardian
		Age	Guardian
		Household income *	Guardian
		Education *	Guardian
		Ethnicity	Guardian
		Marital status *	Guardian
		Gender	Child
		Age *	Child

Note. The asterix denotes cognitive skills or sociodemographic characteristics that were found associated with delay discounting in previous studies. The cognitive skills were reading decoding (Yeh, Myerson, & Green, 2021), language comprehension (Yeh et al., 2021), visuospatial processing (Yeh et al., 2021), and non-verbal reasoning (Shamosh & Gray, 2008; Yeh et al., 2021). The sociodemographics were age (Scheres, Tontsch, Thoeny, & Sumiya, 2014), parental marital status (Sloan, Sanches, Tanabe, & Gowin, 2023), and proxies of SES, such as guardian's household income, education, and ethnicity (Assari, 2020). Gender shows to play no major role for delay discounting behavior in children and adolescent samples where no major disorder is present (Doidge, Flora, & Toplak, 2021)

Table S4*fMRI tasks and their contrasts*

Task	Contrasts	Task description
Stop-Signal Task (SST)	Any stop versus correct go Correct go versus fixation Correct stop versus correct go Correct stop versus incorrect stop Incorrect go versus correct go Incorrect go versus incorrect stop Incorrect stop versus correct go	The SST assesses impulsivity and impulse control. Participants press a button indicating an arrow's direction unless a 'stop' signal appears, requiring them to suppress their response. The stop signal timing adjusts dynamically to maintain a 50% success rate (Casey et al., 2018).
Monetary Incentive Delay Task (MID)	Anticipation of large loss versus neutral Anticipation of large reward versus neutral Anticipation of large versus small loss Anticipation of large reward versus small reward Anticipation of loss versus neutral Anticipation of reward versus neutral Anticipation of small loss versus neutral Anticipation of small reward versus neutral Loss positive versus negative feedback Reward positive versus negative feedback	The MID task examines reward processing and anticipation. Participants see cues indicating potential monetary gains or losses, followed by an anticipation phase and a task to press a button within a time limit. This is followed by feedback. The target display time adjusts automatically to reach an accuracy rate of 60% (Casey et al., 2018).
Emotional nBack Task	0 back versus all other 2 back versus all other 2 back versus 0 back Emotion versus all other Emotion versus neutral face Negative face versus neutral face Positive face versus neutral face Face versus place Place versus all other	The emotional nBack task engages working memory and emotion regulation. Participants see blocks of pictures showing faces with positive, negative or neutral expressions and places. They then determine whether the current image matches a previously shown image, either from the beginning of the block (0-back) or two trials earlier (2-back), varying the cognitive load (Casey et al., 2018).

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